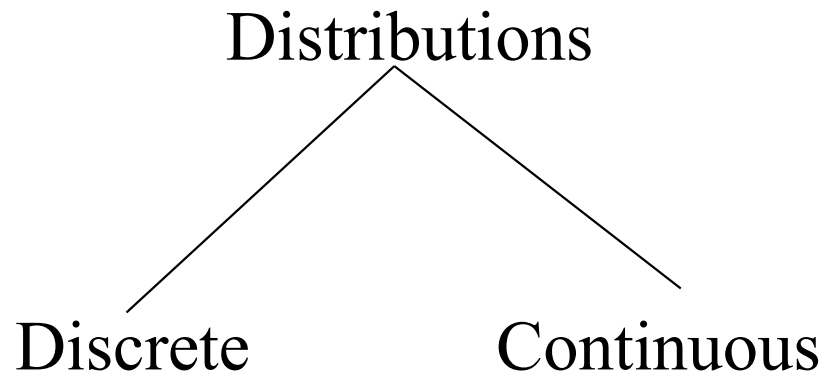


CHAPTER 5

SPECIAL DISCRETE DISTRIBUTIONS

Special Probability Distribution

In this chapter, we shall study some of the probability distributions that figure most prominently in Statistical theory and applications.



1. Discrete Uniform Distribution

If a random variable can take on k different values with equal probability, we say that it has a discrete uniform distribution.

Definition 5.1: A random variable X has a discrete uniform distribution and it is referred to as a discrete uniform random variable if and only if its probability distribution is given by,

$$P(x) = \frac{1}{k} \text{ for } x=x_1, x_2, \dots, x_k \text{ where } x_i \neq x_j \text{ when } i \neq j$$

Examples

Mean of the distribution

$$\mu = E[X] = \sum_{i=1}^k x_i \cdot P(x_i) = \sum_{i=1}^k x_i \cdot \frac{1}{k}$$

Variance of the distribution

$$\sigma^2 = E[(X - \mu)^2] = \sum_{i=1}^k (x_i - \mu)^2 P(x_i)$$
$$\sigma^2 = \sum_{i=1}^k (x_i - \mu)^2 \cdot \frac{1}{k}$$

Example: If X has the discrete uniform distribution

$$P(x) = \frac{1}{k} \text{ for } x=1, 2, \dots, k$$

Show that its mean is

$$\mu = \frac{k+1}{2}$$

•Its variance is

$$\sigma^2 = \frac{(k^2 - 1)}{12}$$

2. Bernoulli Distribution

If an experiment has two outcomes “success” and “failure” and their probabilities are respectively, θ and $(1 - \theta)$, then the number of success 0 or 1 has a Bernoulli distribution.

Definition 5.2: A random variable X has a Bernoulli distribution and it is referred to as a Bernoulli random variable if and only if its probability distribution is given by,

$$P(x; \theta) = \theta^x (1 - \theta)^{(1-x)} ; \text{ for } x=0,1$$

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Thus,

$$P(0; \theta) = (1 - \theta) \quad \text{and} \quad P(1; \theta) = \theta$$

are combined to a single formula.

Example:

A success may be a getting a heads with a balanced coin.

It may be passing (or failing) an examination

It may be the defect (or non defect) status of a component

We refer to an experiment to which the Bernoulli distribution applies as a Bernoulli trial or simply a trial. So, sequences of such experiments as repeated trials.

3. Binomial Distribution

Repeated trials play a very important role in probability and Statistics. Especially when,

- The number of trials is fixed - n
- The parameter θ (the probability of a success) is same for each trial - θ
- The trials are all independent

Several random variables arise in connection with repeated trials. The distribution that we discuss in this section concerns the total number of successes. This has many applications.

- Probability of getting 5 heads in 12 flips.
- Probability that 7 of 10 persons will recover from a disease.

Definition 5.3:

A random variable X has a binomial distribution and it is referred to as a binomial random variable if and only if its probability distribution is given by,

$$P(x; n, \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x} \quad \text{for } x=1,2,\dots,n$$

To derive a formula for probability of getting “x success in n trials”

1-suceess 0-failure

1,0,0,1,1,0,1,0,0,.....,0,1
n trials

If x success then (n-x) failures probability of a success θ , probability of getting x success and n-x failure in specific order is,

$$\theta(1-\theta) (1-\theta) \theta...(1-\theta) \theta = \theta^x(1-\theta)^{n-x}$$

The number of ways in which we can select the x trials on which there is to be a success is $\binom{n}{x}$

and it follows that desired probability for “x success in n trials” is

$$\binom{n}{x} \theta^x (1-\theta)^{n-x}$$